

FRA against ground truth: progress summary

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We built a toy model with planted ground-truth features, validated FRA against it for the first time, then transferred the resulting intervention to the TinyStories sleeper. Four findings bear on the paper, one is a substantial positive result, and one is a clean negative: score-space feature-pair ablation — the operation Figure 1(c) depicts — cannot move a real model at realistic SAE sparsity, for a reason that is arithmetic rather than empirical. Every number traces to a committed artefact in `results/`. Detail in *fra_technical_report*.

1. Things that affect the paper

1.1 The QK pair ranking is position-agnostic. Confidence: high.

`rank_qk_diff` uses $Z_q(1) = \text{sum over ALL query positions of } f[q,1]$. For a sleeper exactly one query matters — the decision position, where the payload is generated. A feature firing forty tokens earlier counts as much as one firing at the decision point.

Feature 1114 fires on 'DE' of DEPLOYMENT, on **0.00%** of clean positions, with mean activation 8.668 on the trigger token against 0.308 elsewhere. It is the trigger detector. Under your mask it ranks **#0, #5, #6 on the query side** — describing attention *from* the trigger — while being live at the decision position in only **3.0%** of deployment rows. The behaviour needs the decision position to attend *to* the trigger, i.e. trigger on the **key** side.

Re-running your function with the only change being the query mask restricted to the final real token (seed-0 `ln1` SAE, your defaults, 200 paired prompts):

	paper mask	decision-position mask
rank-0 pair	(1114, 1232)	(259, 1337)
rank-0 query feature live at decision	3.0%	97.0%
trigger 1114 best as QUERY	#0	#26
trigger 1114 best as KEY	#42	#3
trigger 616 best as KEY	—	#25

Fix: `query_mask = decision_mask`. One argument. The trigger moves to the key side and the query side fills with features live where the decision is made.

This subsumes a frequency bias: because Z is a sum, a feature firing on many positions accumulates a large aggregate wherever those positions are. The rank-0 key feature 1232 fires on 27.72% of deployment and 27.85% of clean positions — not selective at all — while features 616, 1079, 832,

which activate 700–1100x more on the trigger token than elsewhere, never surface. An unweighted position sum rewards how *often* a feature fires and is indifferent to *where*.

1.2 Candidate pairs are not the pairs that were ranked. Confidence: high.

`_top_unique_from_pairs` dedups each side independently; `_get_tuples_diff` then zips **by index**, so the i -th tuple pairs features from different rows of the ranking. The dedup is well motivated — verbatim top-K pairs would give K candidates that are largely one intervention. The zip is the step without a rationale.

tuple	(Q, K)	pair score	rank of that pair, by your own score
0	(1114, 1232)	171.4	#0
1	(1337, 508)	37.16	#27
2	(508, 1241)	4.068	#706
3	(1241, 760)	0.086	#53,421 of 2,359,296
4	(1315, 72)	33.71	#30

Tuple 0 is unaffected, so `--mode winner` is probably fine; the exposure is `--mode topk`, which figure 2 sweeps at `top_k = 20`. **Fix:** take top-K pairs subject to a diversity constraint. A few lines.

1.3 Signed and L1 aggregates coincide within a head. Confidence: proven.

$\text{FRA_QK}[q,k,l,m] = f[q,l] f[k,m] G[l,m]$, and G carries no position indices, so with $f \geq 0$ every term for a fixed (l,m) has the sign of $G[l,m]$. Nothing can cancel, and $|\text{sum}| == \text{sum}|.$ exactly — verified at `max |signed| - L1 | = 0`.

Consequence for note 03: your anti-predictive signed sum (Spearman -0.39 against $+0.86$ for concentration) **cannot** be a within-head effect, since within a head the two aggregates are provably identical up to sign. It is entirely a cross-head and cross-route cancellation phenomenon. That sharpens the result.

It also means “signed and absolute rankings agree” is a tautology, not evidence, whenever the aggregate is over positions within one head.

1.4 Mass fraction is denominator-dominated. Confidence: high.

It divides by an L1 over every co-active pair, and nothing regularises the non-planted entries. In our toy at `rho = 0` the 9,999 non-planted entries carry **96.7%** of the denominator at mean $|G|$ 0.060, against a planted entry of 20.55. So it largely measures how much irrelevant weight the coupling matrix carries; it scales with `d_sae^2` and activation density. **Your 0.074 is not comparable across dictionary sizes.** We report `runner_up_ratio = |planted| / |next largest|` alongside — scale-free, and it moves only when a competitor catches up.

2. Toy model: FRA validated against ground truth

A planted skip-trigram, a 1L/1H attention-only model, W_E frozen at `M @ feature_directions` and $W_{\text{pos}} = 0$ so that `resid_pre == f @ W_dec` exactly. Oracle FRA has a perfect dictionary; no SAE anywhere.

Exactness. $||\text{resid_pre} - \mathbf{f} @ \mathbf{W_dec}|| = 2.98\text{e-}08$; FRA_QK summed over feature pairs reproduces the pre-softmax score to $1.9\text{e-}06$, signed OV likewise. **Both are guaranteed a priori by the construction** — they gate our implementation, they are not findings. We also pass your `tests/fra_conformance/test_synthetic.py`.

Recovery and the rho curve. At $\rho = 0$ the planted pair is **rank 1 of 10,000**, mass fraction 0.0944 against a $1\text{e-}4$ uniform baseline. Sweeping overlap, with held-out accuracy and attention concentration logged as an admission criterion at every point:

rho	held-out	argmax_is_key	circuit		agg mass	G[l*,m*]
			runner-up			
0.00	98.83%	100%	5.83		0.0944	20.55
0.20	99.80%	100%	2.68		0.0200	15.23
0.40	99.80%	100%	2.21		0.0181	14.48
0.80	100.00%	100%	1.54		0.0141	13.61

Margin collapses while the circuit stays perfect, so the degradation is about FRA. What degrades is not the planted coupling (down 34%) but the competitors: the spread of off-diagonal **G** rises **4.9x** and the largest competitor 2.5x. Dispersion, not offset — we checked the obvious rank-1-background explanation and falsified it.

Multi-seed, and it matters. Three seeds per point: 3/3 admitted at $\rho \leq 0.4$, **2/3 at 0.6, 1/3 at 0.8**. Failures are total, not degraded — held-out 9.96–12.01% against a 12.5% chance floor, `argmax_is_key` 1.07–7.62%, **G[l*,m*] negative** (−7.69 to −8.41). So the claim is a conjunction: conditional on the circuit forming, FRA’s margin degrades; and the circuit forms less often as overlap rises. **The defensible range is $\rho \leq 0.4$** . The single-seed curve used seed 0, which was the lucky one.

The circuit-only metric varies by under 2% across seeds where the data-dependent aggregate varies 22–36%. Report the circuit-only curve.

Rank is the wrong metric. Aggregate rank is 1 at *every* ρ , including 0.8 where the runner-up ratio is 1.02 and the edge is effectively tied. A study using only “rank of the planted pair, target 1” would report no degradation at all.

Causal validation. Ablating the planted pair costs **78.81 pp** at $\rho = 0$ (98.83% $\rightarrow$ 20.02%, chance 12.5%); a magnitude-matched random pair costs at most 1.40 pp and the aggregate runner-up at most 0.29 pp. Above $\rho = 0.1$ a unit ablation costs only ~20 pp, but over-ablation collapses behaviour to 0% at every ρ , so that is a magnitude threshold, not redundancy.

Claim A / Claim B — two separable claims, routinely conflated. *Claim A, the site:* a score-row edit perturbs **exactly zero** logits outside the target row — 0.000e+00 on all six (ρ , seed) runs at up to 6x over-ablation — while activation- and residual-space edits perturb 47.7–48.5% of non-target positions. The perturbed set under residual steering is exactly `{positions >= q*}`. This is structural and belongs to **any** score-space intervention; it is not FRA’s. *Claim B, FRA:* FRA is what says which pair to ablate. Ablating an arbitrary pair in score space is equally zero-collateral and equally useless. A claim that FRA-guided steering beats conventional steering asserts both, and only the second is about FRA.

One transferable lesson: our first collateral axis was accuracy at non-target positions and it reported

+0.00 damage for every method at every strength, while residual steering was moving non-target logits by up to 15.16 and perturbing 50.6% of them. Use a distributional measure.

3. The negative result: pair-level intervention does not transfer

The TinyStories sleeper runs on laptop CPU (LoRA merge 11.4s, backdoor fires 6/6 deployment, 0/6 clean). SAEs are not downloadable; we trained one `ln1` seed-0 SAE at your defaults, 1416s. Three arms, selection held fixed per pair, only delivery varying. Suppression of teacher-forced sleeper log-prob at strength 16, baseline -0.073 :

pair	selection	live at dec.	A score-space	B your QK	C ln1 ablate
(1114, 1232)	paper mask, rank 0	3.0%	+0.08	+18.5	+117.2
(259, 1337)	decision mask, rank 0	97.0%	+0.03	+21.2	+140.4
(391, 1114)	decision mask, rank 3	97.0%	−0.01	+8.2	+107.8

Arm A is a null on all three, including both causally-positioned pairs. Fixing selection did not rescue it.

Why: L_0^2 . A (q, k) score decomposes into L_0^2 pair terms. At $L_0 = 32$ that is 1022 terms and a uniform share of 0.0978%. Measured at the decision position against the first key position where the key-side feature fires:

pair	mean share of score	vs uniform
(1114, 1232)	0.326%	3.3x
(391, 1114)	2.144%	21.9x
(259, 1337)	7.201%	73.6x

Identification is working — these are 3–74x the average pair, exactly the concentration FRA is for. They are not causally decisive. Even 7.2% of one cell’s score, over-removed sixteen-fold, does not change what the model generates. In the toy the same object carried **77.9%** of its cell, because L_0 was ~ 4 and the pair was planted. $1/L_0^2$ is the whole difference, and it worsens with k .

Implication. FRA-QK is an attribution tool, not a pair-intervention tool. Figure 1(c) depicts an operation that is well-defined, which we implemented exactly, and which cannot move the model at realistic sparsity. Your actual QK→QK channel sidesteps this by *not* being a pair intervention — it patches `hook_q` with `lambda`’s contribution and `hook_k` with `mu`’s, independently, at each feature’s own firing positions, removing each feature’s contribution to all ~ 32 of its partners. That is L_0 times more score mass than the cross term, and it is why it works. The naming invites a reading the implementation does not support; the implementation is the one that can function.

The toy’s mechanism claim is unaffected: a score-row edit does perturb strictly less than an activation edit. What does not transfer is that a single *pair* is a large enough share to steer with.

4. Open questions for you

1. **Which DGP paper?** Our guess is Chanin and Garriga-Alonso, *Sparse but Wrong* (arXiv:2508.16560). Our generator sits behind a Protocol, so swapping is contained.
2. **Is “feature-matched retrieval” an acceptable task**, or is there a canonical one from the source paper we should use?
3. **Where should the toy live** — this repo or chainik1125/fra?
4. **Does the mean-subtraction trick from the compact-proofs work apply to our G-dispersion problem?** What degrades recovery is the widening *spread* of off-diagonal couplings, not a shift in their mean, and we have not found the right centring. `compute_centered_g` suggests you have thought about a related object.

Settled, so you need not chase it: `autoresearch/cadenza-attn-only` is not the toy experiment — it is the 8B sleeper LoRA study.

5. What is next

Case 2 is untouched and is the open work. The QK→QK no-op confound — KL ~0.1 at the mathematical no-op against ~0.01 for the feature effect — is exactly what a score-space intervention was meant to remove, and `L_0^2` says pair ablation will not do that job at `d_sae = 102k`, `k = 64`, where the share falls another 4x relative to the sleeper SAE. The natural route is score-space ablation of a whole *feature* rather than a pair: remove `sum_mu f[q,1] f[k,mu] G[l,mu]`, which is `L_0` times more mass than the cross term and still never round-trips through the SAE. That is a small change to code that already exists, and it is the experiment we would run next.

Related

- `fra_technical_report` — full detail including background on the method.
- `fra_qk_pair_selection` — findings 1.1 and 1.2 in depth.
- `fra_toy_rho_sweep`, `fra_toy_intervention`, `fra_toy_steering_pareto`, `fra_signed_l1_degeneracy` — the toy results.